YONGSEOP JEONG

Senior Researcher, Electronics and Telecommunications Research Institute (ETRI)

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Robotics · Autonomous Vehicles · Multimodal Sensor Fusion Last updated on September 5, 2026

Experience

Senior Researcher Electronics and Telecommunications Research Institute (ETRI) **Jul. 2023 – Present**

Field Robotics Research Section, AI Robot & Mobility Research Division, Physical AI Research Laboratory

- Navigation AI for a quadrupedal guide-dog robot for the visually impaired (perception, local path planning & tracking, user–robot formation control)

Postdoctoral Researcher Korea Advanced Institute of Science and Technology (KAIST) **Mar. 2023 – Jun. 2023**

Multimodal Sensor Fusion / Autonomous Vehicle System Integration and Implementation

Co-Founder, Director, Pilot Division Lead 3SECONDZ Inc. **Mar. 2019 – Feb. 2023**

Driving-data tech startup (est. 2016) that commercialized the world's first AI-based motorsports coach and built a full driving-data solution — from the in-house data logger "XYRO" to collection, analysis, management and utilization — targeting ADAS, autonomous-driving and connected-car markets.

- Led government-funded R&D projects and exploratory research at the 3SECONDZ R&D Center; self-driving car research & development
- Registered Director (Jun. 2018 – Jun. 2021) · Director of Pangyo R&D Center (Jun. 2020 – Jan. 2022) · Lead of Pilot Division (Jan. 2022 – Feb. 2023)

Education

Doctor of Philosophy Korea Advanced Institute of Science and Technology, Republic of Korea **Feb. 2023**

Robotics, Future Vehicle · Supervisor: In So Kweon

Dissertation: *Purpose-Built Sensor Fusion for Autonomous Vehicles*

Master of Science in Engineering Korea Advanced Institute of Science and Technology, Republic of Korea **Feb. 2013**

Robotics · Supervisor: In So Kweon

Thesis: *An Integrated UGV and UAV Robot System for Effective Surveillance*

Bachelor of Science in Engineering Sungkyunkwan University, Republic of Korea **Feb. 2011**

Electronic and Electrical Engineering · Advisor: Tae-Yong Kuc

Capstone Design: *SODUC – Smart-Oriented Driving in the U-City* (Best Capstone Design Award)

Publication

Full list: Google Scholar

International

Assistive Guidance System Based on Online Path Structure Recognition for the Visually Impaired **2025 Conference**

Jae-Yeong Lee, **Yongseop Jeong**, Seung-Min Choi, Beomsu Seo

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Lightweight Depth Completion Network with Local Similarity-Preserving Knowledge Distillation **2022 Journal**

Yongseop Jeong*, Jinsun Park*, Donghyeon Cho, Yoonjin Hwang, Seibum B. Choi, In So Kweon

Sensors, vol. 22, no. 19, 7388 *Equal contribution

Adaptive Cost Volume Fusion Network for Multi-Modal Depth Estimation in Changing Environments **2022 Journal**

Jinsun Park*, **Yongseop Jeong***, Kyungdon Joo, Donghyeon Cho, In So Kweon

IEEE Robotics and Automation Letters, vol. 7, no. 2, pp. 5095–5102 *Equal contribution

Identification of Vehicle Dynamics Model and Lever-Arm for Arbitrarily Mounted Motion Sensor **2021 Journal**

Yoonjin Hwang, **Yongseop Jeong**, In So Kweon, Seibum B. Choi

IEEE Sensors Journal, vol. 22, no. 10, pp. 9843–9856

Online Misalignment Estimation of Strapdown Navigation for Land Vehicle under Dynamic Condition **2021 Journal**

Yoonjin Hwang, **Yongseop Jeong**, In So Kweon, Seibum B. Choi

International Journal of Automotive Technology, vol. 22, no. 6, pp. 1723–1733

Lane Detection Aided Online Dead Reckoning for GNSS Denied Environments — 2021 Journal
Jinhwan Jeon, Yoonjin Hwang, **Yongseop Jeong**, Sangdon Park, In So Kweon, Seibum B. Choi
Sensors, vol. 21, no. 20, 6805

Visuomotor Understanding for Representation Learning of Driving Scenes — 2019 Conference
Seokju Lee, Junsik Kim, Tae-Hyun Oh, **Yongseop Jeong**, Donggeun Yoo, Stephen Lin, In So Kweon
British Machine Vision Conference (BMVC)

EureCar Turbo: A Self-Driving Car That Can Handle Adverse Weather Conditions — 2016 Conference
Unghui Lee, Jiwon Jung, Seunghak Shin, **Yongseop Jeong**, Kibaek Park, David Hyunchul Shim, In So Kweon
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Relative Pose Estimation for an Integrated UGV-UAV Robot System — 2013 Conference
Yongseop Jeong, In So Kweon
International Conference on Intelligent Robotics and Applications (ICIRA)

DOMESTIC

안내견 로봇 내비게이션 기술 검증을 위한 시뮬레이터 구현 — 2026 Conference
A Simulator for Validating Guide Dog Robot Navigation
정용섭, 장지호, 서범수
ICROS Annual Conference 2026 (제어·로봇·시스템학회 학술대회), pp. 901–902

주행 생성 모델의 학습 효율 가속화를 위한 파라미터 탐색 및 가속화 방법 — 2026 Conference
Exploration and Acceleration Strategies for Efficient Training of Navigation Diffusion Policy
이우주, 카말, 이승익, **정용섭**, 서범수
ICROS Annual Conference 2026 (제어·로봇·시스템학회 학술대회), pp. 1–2

보행로 환경에서 선택적 깊이 추정을 통한 3차원 LiDAR-카메라 융합 객체 검출 기법 — 2025 Conference
Object Detection in Pedestrian Environments Using Selective Depth Estimation with LiDAR-Camera Fusion
임재호, 이우주, **정용섭**, 장지호, 서범수
CEIC 2025 (전자정보통신 학술대회), pp. 1–4

시각장애인 안내견 사족보행 로봇을 위한 지역경로 생성 및 경로 추종 시스템 — 2024 Conference
Local Path Planning and Tracking System for "Guide Dog" Quadrupedal Robot
정용섭, 장지호, 임재호, 최승민, 서범수
CICS 2024 (정보 및 제어 학술대회), pp. 176–177

시각장애인 길 안내 로봇을 위한 멀티 모달 센서 융합 AI 기술 — 2024 Journal
Multi-Modal Sensor Fusion AI Technology for Guide Robots for the Visually Impaired
정용섭
전자공학회지 (The Magazine of the IEIE), Aug. 2024, pp. 23–29

가이드독 로봇과 사용자 간 대형 제어 전략 설계 — 2024 Conference
Design of Formation Control Strategies between User and Guide Dog Robot
정용섭, 나기인, 장지호, 임재호
IEIE Summer Annual Conference 2024 (대한전자공학회 하계학술대회), pp. 3342–3345

가이드독 로봇 모빌리티의 보행 공간 인식과 동적 객체 인식 모듈 설계 및 구현 — 2024 Conference
Design and Implementation of Walking Space Recognition and Dynamic Object Recognition for a Mobility Module of a Guide Dog Robot
나기인, 임재호, 장지호, **정용섭**
IEIE Summer Annual Conference 2024 (대한전자공학회 하계학술대회), pp. 3326–3328

자동차 부품 시험을 위한 자율주행자동차의 활용 — 2022 Conference

Application of Self-Driving Cars for Testing Automotive Parts

정용섭, 이종협, 황윤진

KSAE Fall Conference 2022 (한국자동차공학회 추계학술대회), pp. 890–891

자동차 주행 데이터 수집과 활용 기술 — 2022 Journal

Vehicle Driving Data Collection and Utilization Technology

정용섭

정보과학회지 (Communications of the KIISE), vol. 40, no. 6, pp. 35–40

센서 퓨전을 이용한 자율 주차 시스템 개발 — 2015 Conference

Development of a Self-Parking System Using Sensor Fusion

정지원, 이웅희, 신승학, **정용섭**, 박기백, 심현철, 권인소

KSAE Spring Conference 2015 (한국자동차공학회 춘계학술대회), p. 486

효과적인 감시를 위한 주행 로봇과 비행 로봇의 협동 시스템 — 2013 Conference

An Integrated UGV and UAV Robot System for Effective Surveillance

정용섭, 권인소

ICROS Annual Conference 2013 (제어로봇시스템학회 국내학술대회), pp. 417–418

Project

Guide Dog: Development of Navigation AI Technology of a Guidance Robot for the Visually Impaired Person — Jul. 2023 – Present

Participating Researcher · Lead Organization: Electronics and Telecommunications Research Institute

- Quadrupedal robot navigation: walking-space / dynamic-object recognition, local path planning & tracking, formation control with the user
- Simulator for validating guide-dog robot navigation; assistive guidance based on online path-structure recognition

Lightweight Network based Depth Completion for 3D Object Detection — May 2022 – Dec. 2022

Participating Researcher · Lead Organization: 3SECONDZ Inc.

- Stereo vision + LiDAR + RADAR sensor fusion system implementation for depth completion

Autonomous Vehicle for Tire Endurance Test on Proving Ground — Jul. 2021 – Jun. 2022

Participating Researcher · Lead Organization: 3SECONDZ Inc.

- Converted a mass-produced car into a self-driving car via vehicle CAN
- Sensor fusion system for autonomous driving (vision + GPS + IMU + vehicle stock sensors)
- Performed autonomous tire endurance testing on a proving ground (1,600 km in 3 days)

Active Suspension System for Improvement over 5% Ride and Handling Performance using Road Surface and Road Shape — Apr. 2019 – Dec. 2021

Participating Researcher · Participating Organization: 3SECONDZ Inc.

- Mono-vision-based road surface classifier network design
- Mono-vision-based road shape estimator implementation
- Communication between vehicle ECU and the designed system; system integration with other modules

Award

The 12th Autonomous Vehicle Contest | Hyundai Motor Group — Oct. 2014

Passion Prize (#5), KAIST EureCar Turbo

2014 Student Design Showcase | National Instruments — May 2014

Finalist, EureCar: KAIST Self-Driving Car

The 5th Capstone Design Contest | School of Information and Communication Engineering, Sungkyunkwan University — **Nov. 2010**
Grand Prize (#1), SODUC: Smart-Oriented Driving in U-City

Robot Software Contest | Korea Robotics Society (KROS) — **Aug. 2009**
Silver Prize, ESOP: Easy Operation

National Excellence Scholarship (Science & Engineering) | Korea Student Aid Foundation — **Mar. 2006**
Full Tuition Waiver

MEDIA

'새 눈' 될까..지하철 누비는 AI 안내 로봇 — **Aug. 2026**
MBC 뉴스투데이

[K경제, '20년 궤적'이 만든 '20년 비전'] 2045년 '화성 착륙선' 발사… '20년 뒤' 산업부 기자 구보씨의 일일 — **Nov. 2025**
세계비즈

시각장애인 안내 로봇개 등 ETRI, 미래핵심기술 대공개 — **May 2025**
헤럴드경제

[딥시크, AI 판도 흔든다] 전문가 의견은?… "양날의 검", "국가적 대응 모색" — **Feb. 2025**
세계비즈

"자율주행 개발 나선다" 한국타이어, 스타트업 인수 — **Mar. 2023**
아시아경제

쓰리세컨즈, 가혹환경 주행테스트에 적용 가능한 자율주행기술 세계 최초 상용화 — **Jun. 2022**
매일경제

[쇼미더스포츠] 카레이싱, 3초의 비밀 '코너링' — **Dec. 2018**
채널A 뉴스A

[특집] 2014 자율주행차, 미래를 달려라! — **2014**
OBS경인TV

ACTIVITY

Reviewer
- ETRI Journal (ISSN 1225-6463) — **Dec. 2024 – Present**
- Journal of the Korea Institute of Military Science and Technology (ISSN 1598-9127) — **Oct. 2022 – Present**

Teaching Assistant
- PD513 Future Vehicle Capstone Design, Division of Future Vehicle, KAIST — **Sep. 2013 – Dec. 2016**

REFERENCE

In So Kweon | Physical AI Research Division Director, KIST · iskweon@kaist.ac.kr

Jaewoo Kim | CEO, 3SECONDZ Inc. · jwkim@3secondz.com

Jinsun Park | Associate Professor, Pusan National University +82-51-510-2360 · jspark@pusan.ac.kr

Donghyeon Cho | Associate Professor, Hanyang University +82-2-2220-2657 · doncho@hanyang.ac.kr

Kyungdon Joo | Associate Professor, UNIST +82-52-217-3450 · kyungdon@unist.ac.kr

Soonmin Hwang | Assistant Professor, Hanyang University +82-2-2220-0453 · soonminh@hanyang.ac.kr